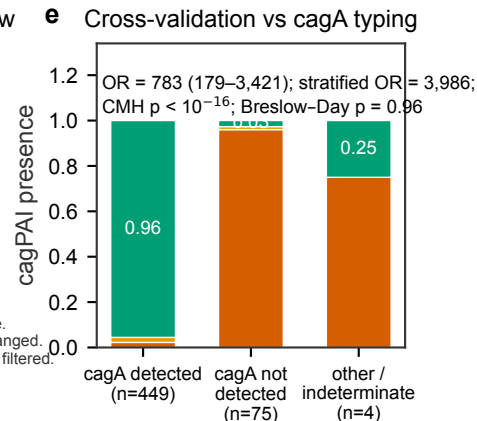
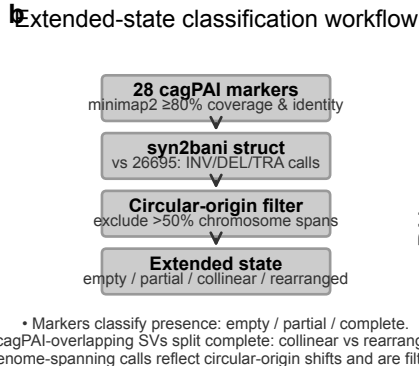
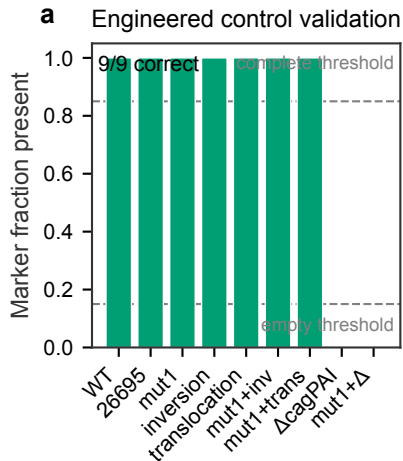
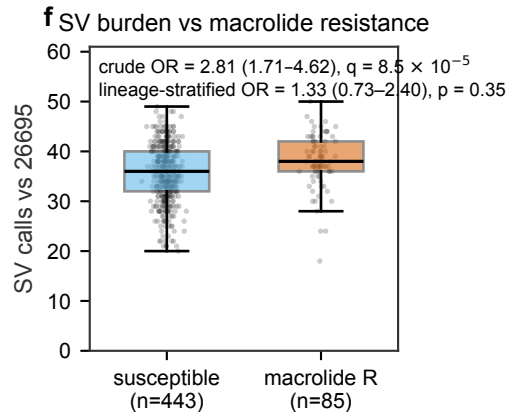
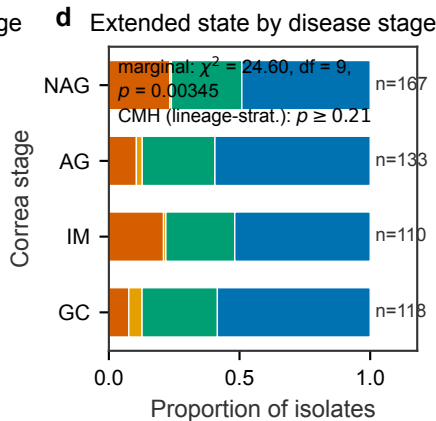
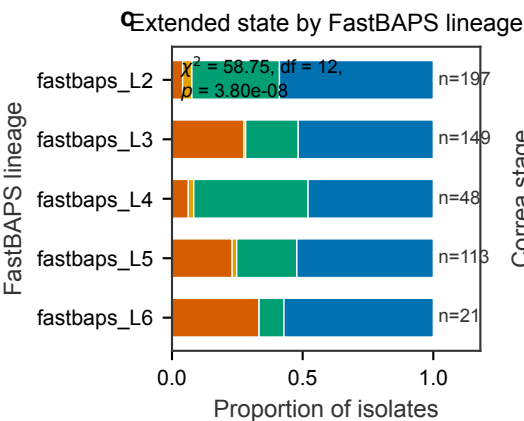




Independent cagA typing (collaborator)



CARD macrolide genotype (23S rRNA)

empty partial complete collinear complete rearranged